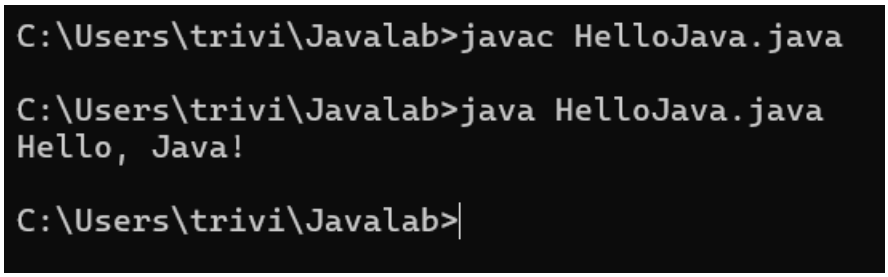
1. **Write a Java program to display "Hello, Java!" and demonstrate the basic structure of a Java program.**

```
public class HelloJava {
    public static void main(String[] args) {
        System.out.println("Hello, Java!");
    }
}
```



2. **Write a Java program to declare variables of different primitive data types (byte, short, int, long, float, double, char, and boolean) and display their values.**

```
public class PrimitiveDataTypes {
    public static void main(String[] args) {
        byte b = 100;
        short s = 20000;
        int i = 500000;
        long l = 9000000000L;
        float f = 12.5f;
        double d = 99.999;
        char c = 'A';
        boolean a = true;

        System.out.println("Byte: " + b);
```

```
        System.out.println("Short: " + s);
        System.out.println("Int: " + i);
        System.out.println("Long: " + l);
        System.out.println("Float: " + f);
        System.out.println("Double: " + d);
        System.out.println("Char: " + c);
        System.out.println("Boolean: " + a);
    }
}
```

```
C:\Users\trivi\Javalab>javac PrimitiveDataTypes.java

C:\Users\trivi\Javalab>java PrimitiveDataTypes.java
Byte: 100
Short: 20000
Int: 500000
Long: 9000000000
Float: 12.5
Double: 99.999
Char: A
Boolean: true

C:\Users\trivi\Javalab>
```

3. **Write a Java program to perform arithmetic operations (addition, subtraction, multiplication, division, and modulus) on two integer variables.**

```
public class Arthmtc {
    public static void main(String[] args) {
        int a = 20;
        int b = 6;
        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
```

```
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));
    }
}
```

```
C:\Users\trivi\Javalab>javac Arthmtc.java

C:\Users\trivi\Javalab>java Arthmtc.java
Addition: 26
Subtraction: 14
Multiplication: 120
Division: 3
Modulus: 2

C:\Users\trivi\Javalab>
```

4. **Write a Java program to perform arithmetic operations on floating-point numbers and display the results.**

```
public class Float {
    public static void main(String[] args) {
        float a = 15.5f;
        float b = 4.5f;

        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));
    }
}
```

```
C:\Users\trivi\Javalab>javac Float.java

C:\Users\trivi\Javalab>java Float.java
Addition: 20.0
Subtraction: 11.0
Multiplication: 69.75
Division: 3.4444444
Modulus: 2.0

C:\Users\trivi\Javalab>
```

5. **Write a Java program to demonstrate character data type by displaying a character and its corresponding ASCII/Unicode value.**

```
public class CharacterDemo {
    public static void main(String[] args) {
        char ch = 'A';


        System.out.println("Character: " + ch);
        System.out.println("ASCII/Unicode Value: " + (int) ch);
    }
}
```

```
C:\Users\trivi\Javalab>javac CharacterDemo.java

C:\Users\trivi\Javalab>java CharacterDemo.java
Character: A
ASCII/Unicode Value: 65

C:\Users\trivi\Javalab>
```

6. **Write a Java program to demonstrate the use of Boolean variables and Boolean expressions.**

```
public class Bool {
    public static void main(String[] args) {
        boolean isJavaFun = true;
```

```
        boolean isGreater = 15 > 10;


        System.out.println("Is Java Fun? " + isJavaFun);
        System.out.println("Is 15 greater than 10? " + isGreater);
    }
}
```

```
C:\Users\trivi\Javalab>javac Bool.java

C:\Users\trivi\Javalab>java Bool.java
Is Java Fun? true
Is 15 greater than 10? true

C:\Users\trivi\Javalab>
```

7. **Write a Java program to declare and initialize different types of variables and display their values.**

```
public class VariableDemo {
    public static void main(String[] args) {
        int age = 19;
        double cgpa = 9;
        char grade = 'A';
        boolean isStudent = true;
        String name = "Aditya";


        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("CGPA: " + cgpa);
        System.out.println("Grade: " + grade);
        System.out.println("Student: " + isStudent);
    }
```

```
}
```

```
C:\Users\trivi\Javalab>javac VariableDemo.java

C:\Users\trivi\Javalab>java VariableDemo.java
Name: Aditya
Age: 19
CGPA: 9.0
Grade: A
Student: true

C:\Users\trivi\Javalab>
```

8. **Write a Java program to swap the values of two variables using a temporary variable.**

```
public class Swap {
    public static void main(String[] args) {
        int a = 10;
        int b = 20;
        int temp;

        System.out.println("Before Swapping:");
        System.out.println("a = " + a);
        System.out.println("b = " + b);

        temp = a;
        a = b;
        b = temp;

        System.out.println("After Swapping:");
        System.out.println("a = " + a);
        System.out.println("b = " + b);
    }
```

}

```
C:\Users\trivi\Javalab>javac Swap.java

C:\Users\trivi\Javalab>java Swap.java
Before Swapping:
a = 10
b = 20
After Swapping:
a = 20
b = 10

C:\Users\trivi\Javalab>
```

9. **Write a Java program to demonstrate automatic (widening) type conversion between primitive data types.**

```
public class Widening {
    public static void main(String[] args) {
        int num = 50;
        double result = num;

        System.out.println("Integer Value: " + num);
        System.out.println("Double Value: " + result);
    }
}
```

```
C:\Users\trivi\Javalab>javac Widening.java

C:\Users\trivi\Javalab>java Widening.java
Integer Value: 50
Double Value: 50.0

C:\Users\trivi\Javalab>
```

10. **Write a Java program to demonstrate explicit (narrowing) type casting from one primitive data type to another.**

```
public class Narrowing {
```

```
    public static void main(String[] args) {
        double num = 25.75;
        int result = (int) num;

        System.out.println("Double Value: " + num);
        System.out.println("Integer Value: " + result);
    }
}
```

```
C:\Users\trivi\Javalab>javac Narrowing.java

C:\Users\trivi\Javalab>java Narrowing.java
Double Value: 25.75
Integer Value: 25

C:\Users\trivi\Javalab>
```

11. Write a Java program to convert a character to its ASCII/Unicode value and vice versa using type casting.

```
public class CharConv{
    public static void main(String[] args) {
        char ch = 'A';
        int ascii = (int) ch;

        int value = 66;
        char letter = (char) value;

        System.out.println("Character: " + ch);
        System.out.println("ASCII/Unicode Value: " + ascii);
        System.out.println("ASCII/Unicode Value: " + value);
        System.out.println("Character: " + letter);
```

```
    }
}
```

```
C:\Users\trivi\Javalab>javac CharConv.java

C:\Users\trivi\Javalab>java CharConv.java
Character: A
ASCII/Unicode Value: 65
ASCII/Unicode Value: 66
Character: B

C:\Users\trivi\Javalab>
```

12. Write a Java program to check whether a given number is even or odd using the if-else control statement.

```
public class EvenOdd {
    public static void main(String[] args) {
        int num=57;
       if (num % 2 == 0) {
          System.out.println(num + " is Even");
       } else {
          System.out.println(num + " is Odd");
       }

    }
}
```

```
C:\Users\trivi\Javalab>javac EvenOdd.java

C:\Users\trivi\Javalab>java EvenOdd.java
57 is Odd

C:\Users\trivi\Javalab>
```

13. Write a Java program to find the largest of two numbers using the if-else statement.

```
public class Lrgst {
    public static void main(String[] args) {
        int num1 = 25;
        int num2 = 18;

        if (num1 > num2) {
            System.out.println(num1 + " is the largest number.");
        } else if (num2 > num1) {
            System.out.println(num2 + " is the largest number.");
        } else {
            System.out.println("Both numbers are equal.");
        }
    }
}
```

```
C:\Users\DELL\OneDrive\Desktop\25EU02068>javac Lrgst.java

C:\Users\DELL\OneDrive\Desktop\25EU02068>java Lrgst.java
25 is the largest number.

C:\Users\DELL\OneDrive\Desktop\25EU02068>
```

14. Write a Java program to demonstrate the use of Java reserved keywords by creating valid identifiers and explaining why keywords cannot be used as variable names.

```
public class KeywordsDemo {
    public static void main(String[] args) {
        int age = 19;
        double salary = 50000.0;
        String studentName = "Harini";
```

```
        System.out.println("Age: " + age);
        System.out.println("Salary: " + salary);
        System.out.println("Student Name: " + studentName);

        System.out.println("Keywords like int, class, if, and while cannot be used as variable names.");
    }
}
```

```
C:\Users\trivi\Javalab>javac KeywordsDemo.java

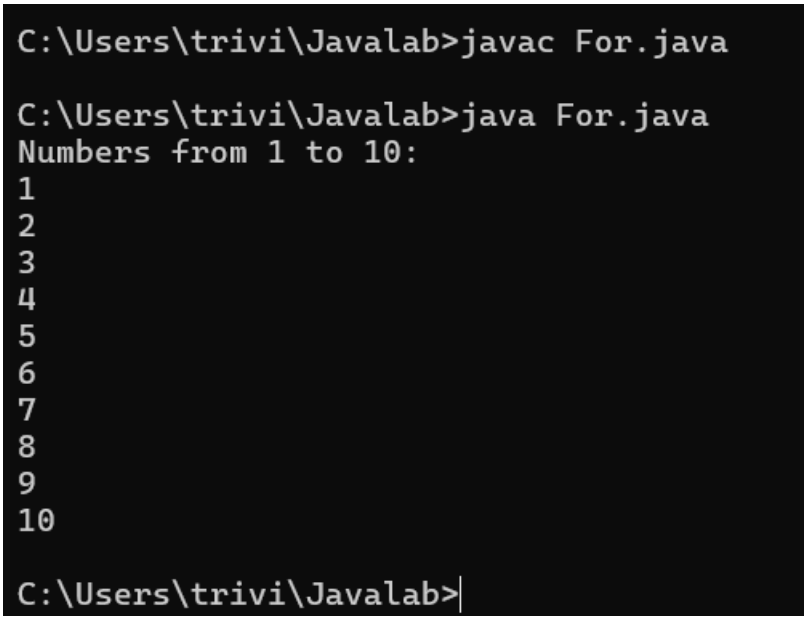
C:\Users\trivi\Javalab>java KeywordsDemo.java
Age: 19
Salary: 100000.0
Student Name: Aditya
Keywords like int, class, if, and while cannot be used as variable names.

C:\Users\trivi\Javalab>
```

15. Write a Java program to demonstrate the use of the for loop by displaying numbers from 1 to 10.

```
public class For {
    public static void main(String[] args) {
        System.out.println("Numbers from 1 to 10:");

        for (int i = 1; i <= 10; i++) {
            System.out.println(i);
        }
    }
}
```


C:\Users\trivi\Javalab>javac For.java
C:\Users\trivi\Javalab>java For.java
Numbers from 1 to 10:
1
2
3
4
5
6
7
8
9
10
C:\Users\trivi\Javalab>